Mohamed Jubair K

Full Stack Developer — React.js — Node.js — Express.js — MongoDB — JavaScript — Python

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India — Target: Full Stack, MERN Stack, Software Engineering, Web Development — English, Tamil

Professional Summary

B.Tech Artificial Intelligence & Data Science graduate targeting Full Stack Developer roles, with hands-on experience in MERN stack development, REST APIs, JWT authentication, RBAC, database optimization, and responsive web applications. Skilled in React.js, Next.js, Node.js, Express.js, MongoDB, PostgreSQL, JavaScript, Python, Git, Docker, cloud deployment, and AI/ML concepts including PyTorch, TensorFlow, NLP, LLMs, RAG, and computer vision.

Technical Skills

Languages: JavaScript, Python, SQL, Java, C++
Full Stack: React.js, Next.js, Node.js, Express.js, REST APIs, JWT, RBAC, Responsive Web Design
Databases: MongoDB, PostgreSQL, Firebase, Firestore, SQLite, Query Optimization
DevOps & Tools: Docker, Kubernetes, AWS EC2, AWS S3, Git, GitHub, Linux, Vercel, VS Code, Postman, Jupyter
AI/ML & Data: PyTorch, TensorFlow, Scikit-learn, Hugging Face, Transformers, CNN, RNN, LLMs, RAG, Prompt Engineering, Pandas, NumPy, Power BI
Core CS: DSA, OOP, DBMS, Operating Systems, Computer Networks, Microservices, Kafka, RabbitMQ, Redis, Load Balancing

Experience

Full Stack Developer – Coding Contest Platform — **2025**

React.js, Node.js, MongoDB, REST APIs

- Built a production-ready coding contest platform with REST APIs, JWT authentication, RBAC, real-time leaderboard, and automated code evaluation.
- Implemented backend workflows using Node.js, Express.js, and MongoDB for contests, submissions, rankings, and user access control.
- Improved database performance through MongoDB query optimization for contest and leaderboard workflows.

AI/ML Research Intern – Self-Initiated Research Projects — **2024–Present**

AI Research, Deep Learning, Computer Vision, Embedded Systems

- Researched restricted self-attention mechanisms for transformer complexity reduction and inference optimization.
- Developed experiments achieving up to 40% faster inference and reduced memory usage in transformer workflows.
- Built multimodal species recognition using vision, audio, and text inputs, achieving 92% accuracy.

Projects

Coding Contest Platform — GitHub

Full Stack Web Platform, MERN Stack, Backend APIs

- Created a production-ready coding platform with authentication, RBAC, automated code evaluation, and real-time ranking.
- Designed REST API workflows using Node.js, Express.js, MongoDB, and JWT-based authentication.
- Optimized MongoDB queries for contest, submission, and leaderboard performance.

Tech Stack: React.js, Node.js, Express.js, MongoDB, JWT, REST APIs

Efficient Transformer with Restricted Self-Attention — GitHub

Transformer Optimization, Model Efficiency, Research Project

- Designed a restricted self-attention mechanism for transformer optimization and complexity reduction.
- Achieved 35% faster inference and 40% lower memory usage in model experiments.
- Presented at IIT Madras Technical Symposium and won Third Prize.

Tech Stack: PyTorch, CUDA, Hugging Face, NumPy

Multimodal Species Recognition — GitHub

Computer Vision, Audio Processing, Multimodal AI

- Built a multimodal AI system combining image, audio, and text features for species classification.
- Achieved 92% accuracy across multimodal recognition workflows and packaged the solution using Docker.

Tech Stack: PyTorch, OpenCV, Librosa, Docker

Achievements

- 2nd Prize, National Paper Presentation, MRK College.
- 3rd Prize, IIT Madras Technical Symposium.
- Presented research in 5+ technical symposiums.
- AI/ML Open Source Contributor.

Certifications

NPTEL Cloud Computing – Elite — IBM Artificial Intelligence — Great Learning UI/UX Design

Education

C.K. College of Engineering and Technology — **2022–2026**

B.Tech in Artificial Intelligence & Data Science — CGPA: 8.5/10